Vasicek vs CR+

Jan Alsterlind

July 2026

1 Introduction

The Vasicek model and CreditRisk+ (CR+) are two of the most widely used frameworks for modelling credit portfolio risk. Although their mathematical formulations differ substantially, both models are designed to describe the same economic phenomenon: the impact of systematic credit risk on a large portfolio of obligors. In both frameworks, defaults become correlated because borrowers are exposed to a common risk driver representing the state of the economy. In the asymptotic limit of a large homogeneous portfolio, idiosyncratic risk is diversified away and portfolio losses are entirely determined by this common factor. Consequently, both models can be viewed as asymptotic single-factor credit portfolio models. In the large portfolio limit, portfolio losses are driven primarily by a single systematic risk factor. The principal difference between the models lies in how the systematic risk factor is represented:

- In the Vasicek model, systematic risk is described by a latent normally distributed factor Z and an asset correlation parameter ρ.
- In CreditRisk+, systematic risk is represented by a stochastic default intensity Λ, typically assumed to follow a gamma distribution.

Despite these differing representations, the models can be calibrated so that their loss distributions become nearly identical in the large portfolio limit. This can be achieved either by matching the variance of the common risk driver or by matching a specific quantile of the loss distribution, such as the 99% Value-at-Risk. Both approaches are discussed below.

2 Systematic Risk in Vasicek and CreditRisk+

Both models generate default dependence through a portfolio-wide systematic state variable. The specific mathematical constructions differ, but the economic interpretation is the same: favourable systematic states produce few defaults, while adverse states produce large clusters of defaults.

2.1 Vasicek Model

In the Vasicek model, the credit quality of borrower i is represented by the latent variable

$$X_i = \sqrt{\rho}\, Z + \sqrt{1-\rho}\, \varepsilon_i,$$

where

$$Z \sim N(0,1), \qquad \varepsilon_i \sim N(0,1),$$

and the idiosyncratic variables ε_i are mutually independent and independent of Z. A default occurs when

$$X_i < \Phi^{-1}(p),$$

where p denotes the unconditional probability of default. Conditioning on a realization $Z = z$ gives

$$P(X_i < \Phi^{-1}(p) \mid Z = z) = P\left(\sqrt{\rho}\, z + \sqrt{1-\rho}\, \varepsilon_i < \Phi^{-1}(p)\right).$$

Rearranging yields

$$P\left(\varepsilon_i < \frac{\Phi^{-1}(p) - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right).$$

Since ε_i is standard normally distributed,

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right).$$

This conditional probability of default is the key quantity governing portfolio losses in the asymptotic limit. A large positive value of z corresponds to a strong economy and a low conditional default rate, whereas a large negative value of z produces elevated default probabilities and clustered losses.

2.2 CreditRisk+

CreditRisk+ introduces systematic dependence through a stochastic default intensity. Let N denote the number of defaults in a portfolio during a given time horizon. Conditional on a realization of the intensity factor $\Lambda = \lambda$, defaults are assumed to occur according to a Poisson distribution,

$$N \mid \Lambda = \lambda \sim \text{Poisson}(\lambda).$$

It follows that

$$E[N \mid \Lambda = \lambda] = \lambda, \qquad \text{Var}(N \mid \Lambda = \lambda) = \lambda.$$

Systematic risk is introduced by allowing the default intensity itself to vary across economic states. In the standard CreditRisk+ formulation,

$$\Lambda \sim \text{Gamma}(\alpha, \beta),$$

where α denotes the shape parameter and β the scale parameter. The moments of the gamma distribution are

$$E[\Lambda] = \alpha\beta,$$

and

$$\text{Var}(\Lambda) = \alpha\beta^2.$$

The mean determines the unconditional expected number of defaults, while the variance determines the degree of systematic uncertainty around that expectation. A useful measure of the variability of the common risk factor is the squared coefficient of variation,

$$CV^2 = \frac{\text{Var}(\Lambda)}{E[\Lambda]^2}.$$

Substituting the moments of the gamma distribution gives

$$CV^2 = \frac{\alpha\beta^2}{(\alpha\beta)^2},$$

which simplifies to

$$CV^2 = \frac{1}{\alpha}.$$

Hence, the shape parameter completely determines the relative variability of the common intensity factor. A low value of α corresponds to a highly volatile credit environment, whereas a large value of α produces a more stable environment with weaker systematic fluctuations. The economic interpretation is analogous to the Vasicek model. Different realizations of Λ correspond to different systematic states of the economy: a low value of Λ implies a benign credit environment with relatively few defaults, while a high value of Λ corresponds to stressed conditions with elevated default rates throughout the portfolio.

2.3 Common Interpretation

The two models can therefore be viewed as alternative representations of the same underlying mechanism. In the Vasicek model, the common state variable is the Gaussian factor Z, which determines the conditional probability of default

$$p(Z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\,Z}{\sqrt{1-\rho}}\right).$$

In CreditRisk+, the common state variable is the gamma-distributed intensity

$$\Lambda,$$

which directly determines the conditional default frequency. In the large portfolio limit, portfolio losses are fully determined by the realization of these common factors. The asymptotic loss rate may therefore be written schematically as

$$L_\infty = p(Z)$$

in the Vasicek model and

$$L_\infty \propto \Lambda$$

in CreditRisk+. To illustrate the similarity, consider a portfolio of 10 000 identical loans with an unconditional probability of default of 1%. The expected number of defaults is therefore 100 per year. In CreditRisk+, different realizations of the common intensity correspond to different economic states:

- A favourable state with a low realization of Λ, producing substantially fewer than 100 defaults.
- An average state with $\Lambda \approx E[\Lambda]$, producing approximately 100 defaults.
- An adverse state with a high realization of Λ, producing a large cluster of defaults.

Exactly the same interpretation can be made in the Vasicek model. Once the systematic factor takes a realized value $Z = z$, all borrowers share the same conditional probability of default

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\,z}{\sqrt{1-\rho}}\right).$$

The realization of z therefore determines the overall state of the credit portfolio:

- A favourable state corresponds to a high value of z, for which $p(z)$ becomes substantially lower than 1% and only a small number of loans default.
- An average state corresponds approximately to $z \approx 0$, for which the default rate is close to its unconditional level and the number of defaults is around 100.
- An adverse state corresponds to a low value of z, for which $p(z)$ increases significantly and a large cluster of defaults occurs.

In both models, defaults are conditionally independent once the common risk factor has been fixed. In CreditRisk+, this factor is the stochastic intensity Λ, while in the Vasicek model it is the Gaussian factor Z. Default clustering therefore does not arise from direct interactions between obligors, but from their shared exposure to the same systematic state of the economy. From this perspective, the principal difference between the models is not the economic mechanism but the mathematical representation of the systematic risk driver. Vasicek models the state of the economy through a normally distributed latent factor, whereas CreditRisk+ models it through a gamma-distributed default intensity. Both approaches generate a single-factor loss distribution whose shape is determined by the variability of the common systematic risk factor.

2.4 Differences in Tail Behaviour

Although the Vasicek model and CreditRisk+ represent the same economic mechanism of systematic default clustering, they do not generally produce identical loss distributions. The most important difference arises from the way the systematic factor is transformed into portfolio losses. In CreditRisk+, the systematic factor enters linearly through the stochastic default intensity

$$L_\infty \propto \Lambda,$$

where Λ is gamma distributed. Consequently, large portfolio losses are generated directly by large realizations of the intensity factor. The shape of the upper tail therefore reflects the tail properties of the gamma distribution itself. In the Vasicek model, the systematic factor enters through the nonlinear transformation

$$L_\infty = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, Z}{\sqrt{1-\rho}}\right).$$

A sufficiently adverse realization of the Gaussian factor Z is mapped into a conditional probability of default that can be many times larger than the unconditional probability of default. Consequently, the relationship between the systematic factor and the portfolio loss is highly nonlinear. This distinction has important implications for the tail of the loss distribution. At first sight, one might expect CreditRisk+ to produce the heavier tail because the gamma distribution is right-skewed and exhibits positive excess kurtosis, whereas the Vasicek factor is Gaussian. However, the nonlinear probit transformation in the Vasicek model amplifies adverse systematic shocks. As a result, the Vasicek loss distribution can exhibit equal or even greater tail losses than CreditRisk+ at very high confidence levels, despite being driven by a normally distributed factor. The practical implication is that matching lower-order moments, such as the variance of the systematic factor, does not guarantee agreement in extreme tail quantiles. Two models may be calibrated to have the same expected loss and a similar variance while still producing materially different values of economic capital or stress losses. Consequently, it is often necessary to calibrate directly to tail quantiles when the objective is to obtain close agreement in the economically relevant part of the loss distribution.

3 Approximate Mapping between Vasicek Asset Correlation and the Variability of the CreditRisk+ Systematic Factor

Although the Vasicek model and CreditRisk+ are based on different mathematical frameworks, both separate credit risk into an idiosyncratic component and a systematic component driven by a common economic factor. In the Vasicek model, this decomposition is explicit in the latent asset variable. In CreditRisk+, it appears through the decomposition of default-count variance into ordinary Poisson variation and variation in a stochastic default intensity. This common structure makes it possible to compare the two models by comparing the relative importance of systematic and idiosyncratic risk. Doing so leads to a useful approximation linking the Vasicek asset correlation ρ to the CreditRisk+ gamma volatility.

3.1 Systematic and idiosyncratic risk in the Vasicek model

In the Vasicek model, the latent variable of obligor i is

$$X_i = \sqrt{\rho}\, Z + \sqrt{1-\rho}\, \varepsilon_i,$$

where both Z and ε_i are standard normally distributed and mutually independent. Since

$$\mathrm{Var}(X_i) = \rho + (1-\rho) = 1,$$

the total variance can be decomposed into

$$\underbrace{\rho}_{\text{systematic risk}} + \underbrace{(1-\rho)}_{\text{idiosyncratic risk}}.$$

The parameter ρ therefore represents the fraction of total asset variance attributable to the common systematic factor, while $(1-\rho)$ represents the fraction attributable to borrower-specific effects. A natural measure of the relative importance of systematic risk is therefore

$$\frac{\text{systematic risk}}{\text{idiosyncratic risk}} = \frac{\rho}{1-\rho}.$$

When ρ is small, defaults are driven primarily by borrower-specific events and portfolio losses exhibit little dependence on the economic environment. As ρ increases, the common factor becomes more important and defaults become increasingly clustered during adverse economic conditions. In the asymptotic portfolio limit, idiosyncratic fluctuations are diversified away and portfolio losses become entirely driven by the realization of the common factor Z. The ratio

$$\frac{\rho}{1-\rho}$$

can therefore be viewed as a natural measure of the strength of systematic credit risk in the Vasicek framework.

3.2 Systematic and idiosyncratic risk in CreditRisk+

In CreditRisk+, the number of defaults is assumed to be conditionally Poisson distributed,

$$N \mid \Lambda \sim \mathrm{Poisson}(\Lambda),$$

where the default intensity Λ follows a gamma distribution. Applying the law of total variance gives

$$\mathrm{Var}(N) = E[\mathrm{Var}(N \mid \Lambda)] + \mathrm{Var}(E[N \mid \Lambda]).$$

Since

$$E[N \mid \Lambda] = \Lambda, \qquad \mathrm{Var}(N \mid \Lambda) = \Lambda,$$

it follows that

$$\mathrm{Var}(N) = E[\Lambda] + \mathrm{Var}(\Lambda).$$

This decomposition can be interpreted as

$$\mathrm{Var}(N) = \underbrace{E[\Lambda]}_{\text{idiosyncratic risk}} + \underbrace{\mathrm{Var}(\Lambda)}_{\text{systematic risk}}.$$

The first term represents ordinary Poisson variation and corresponds to borrower-specific randomness. The second term captures fluctuations in the common default intensity and therefore represents systematic risk. By analogy with the Vasicek model, a natural measure of the relative importance of systematic risk would therefore be

$$\frac{\text{systematic risk}}{\text{idiosyncratic risk}} = \frac{\mathrm{Var}(\Lambda)}{E[\Lambda]}.$$

However, this quantity depends on the scale of the portfolio through $E[\Lambda]$. Doubling the portfolio size doubles the expected number of defaults and changes the ratio, even if the underlying systematic risk remains unchanged. A more useful measure is therefore obtained by normalizing the stochastic intensity by its mean. For a gamma-distributed intensity,

$$CV^2 = \frac{\mathrm{Var}(\Lambda)}{E[\Lambda]^2},$$

where CV denotes the coefficient of variation. The quantity CV^2 is dimensionless and independent of portfolio size. It measures the variability of the common credit environment relative to its average level. A larger value of CV^2 implies larger fluctuations in the common default intensity and consequently stronger dependence between obligors. In this sense, CV^2 plays a role analogous to

$$\frac{\rho}{1-\rho},$$

which measures the relative importance of the common systematic factor in the Vasicek model.

3.3 Approximate parameter mapping

Both models can therefore be interpreted as decomposing credit risk into an idiosyncratic component and a systematic component. In the Vasicek model, the relative importance of these components is measured by

$$\frac{\rho}{1-\rho},$$

whereas in CreditRisk+ the variability of the common risk factor is measured by

$$CV^2.$$

Identifying these measures of systematic risk gives the approximation

$$CV^2 \approx \frac{\rho}{1-\rho}.$$

Solving for ρ yields

$$\rho \approx \frac{CV^2}{1+CV^2}.$$

This relationship provides a practical heuristic mapping between the CreditRisk+ gamma volatility and the Vasicek asset correlation. While not exact, it captures the idea that both parameters quantify the strength of systematic credit risk relative to idiosyncratic risk.

4 Calibrations

Throughout this section, the unconditional probability of default p is assumed to be known and identical in both models. The CreditRisk+ model is treated as the reference specification, while the Vasicek model is calibrated to reproduce selected characteristics of the CreditRisk+ loss distribution. Consequently, the calibration problem is not to estimate both models jointly. Instead, the parameters governing the CreditRisk+ systematic risk factor Λ are assumed to be known, and we determine the asset correlation parameter ρ in the Vasicek model that yields the closest correspondence between the two frameworks. Three calibration approaches are considered. The first matches the variance of the systematic risk factor, the second matches a single tail quantile exactly, and the third matches several tail quantiles simultaneously in a least-squares sense.

4.1 Calibration via gamma variance and asset correlation

Assume that the asset correlation in the Vasicek model is $\rho = 10\%$. At a 99% confidence level, the outcome corresponds to a strongly negative realization of the systematic factor Z. The conditional PD then becomes significantly higher than 1%, which leads to a large cluster of defaults. By exploiting the relationship derived earlier, we translate this into CreditRisk+ using

$$CV^2 \approx \frac{\rho}{1-\rho}.$$

In the numerical example,

$$\rho = 0.10.$$

Inserted into the expression above yields

$$CV^2 \approx \frac{0.10}{1-0.10} = \frac{0.10}{0.90} \approx 0.111.$$

By definition, the coefficient of variation satisfies

$$CV = \frac{\sqrt{\mathrm{Var}(\Lambda)}}{E[\Lambda]}.$$

Thus,

$$CV = \sqrt{CV^2} = \sqrt{0.111} \approx 0.333.$$

This calibration is obtained by matching the variability of the systematic risk factor and should not be interpreted as an exact matching of any particular loss quantile.

Economic interpretation This implies that the standard deviation of the common default intensity Λ is approximately 33% of its mean. At the 99% quantile, a high value of Λ is realized, leading to essentially the same number of defaults as in the Vasicek model. Numerical comparisons typically show that the 99% VaR differs only marginally between the two models.

4.2 Calibration of ρ to a Given VaR Level

In practical applications, it may be desirable not only to match the variance of the common risk driver but also to match a specific quantile of the loss distribution. This is particularly relevant when CreditRisk+ is used as the primary model and one wishes to determine the corresponding asset correlation in the Vasicek framework. Assume that the 99% loss quantile generated by CreditRisk+ is

$$q_{CR+}(0.99).$$

We seek the value of ρ such that the Vasicek model produces the same loss quantile:

$$q_V(0.99;\rho) = q_{CR+}(0.99).$$

Solution for ρ To match the CreditRisk+ 99% loss quantile, we solve

$$\Phi\left(\frac{\Phi^{-1}(p) + \sqrt{\rho}\,\Phi^{-1}(0.99)}{\sqrt{1-\rho}}\right) = q_{CR+}(0.99).$$

Applying the inverse standard normal distribution function to both sides gives

$$\frac{\Phi^{-1}(p) + \sqrt{\rho}\,\Phi^{-1}(0.99)}{\sqrt{1-\rho}} = \Phi^{-1}(q_{CR+}(0.99))\,.$$

Define

$$a = \Phi^{-1}(p), \qquad b = \Phi^{-1}(0.99), \qquad c = \Phi^{-1}(q_{CR+}(0.99))\,.$$

The calibration condition becomes

$$\frac{a + b\sqrt{\rho}}{\sqrt{1-\rho}} = c.$$

To eliminate the square roots, let

$$x = \sqrt{\rho}.$$

Since $0 \leq \rho \leq 1$, it follows that $0 \leq x \leq 1$. The equation becomes

$$\frac{a + bx}{\sqrt{1-x^2}} = c.$$

Multiplying both sides by $\sqrt{1-x^2}$ and squaring yields

$$(a + bx)^2 = c^2(1 - x^2).$$

Expanding and collecting terms gives the quadratic equation

$$(b^2 + c^2)x^2 + 2ab\,x + (a^2 - c^2) = 0.$$

The solutions are therefore

$$x = \frac{-2ab \pm \sqrt{(2ab)^2 - 4(b^2+c^2)(a^2-c^2)}}{2(b^2+c^2)}.$$

Since $x = \sqrt{\rho} \geq 0$, the economically relevant root is the non-negative solution. Once x has been determined, the implied asset correlation follows from

$$\rho = x^2.$$

Because the Vasicek loss quantile is a monotonic increasing function of the asset correlation, the solution is unique.

Numerical example Suppose that the portfolio PD is

$$p = 1\% = 0.01,$$

and that CreditRisk+ produces a 99%-VaR of

$$q_{CR+}(0.99) = 5\% = 0.05.$$

Then

$$a = \Phi^{-1}(0.01) \approx -2.3263,$$
$$b = \Phi^{-1}(0.99) \approx 2.3263,$$

and

$$c = \Phi^{-1}(q_{CR+}(0.99)) = \Phi^{-1}(0.05) \approx -1.6449.$$

Substituting into the quadratic equation yields

$$(2.3263^2 + 1.6449^2)x^2 + 2(-2.3263)(2.3263)x + (2.3263^2 - 1.6449^2) = 0,$$

or

$$8.1174x^2 - 10.8238x + 2.7064 = 0.$$

Applying the quadratic formula gives

$$x_1 \approx 1.000, \qquad x_2 \approx 0.333.$$

The relevant solution is

$$x = \sqrt{\rho} \approx 0.333,$$

and hence

$$\rho = x^2 \approx 0.111.$$

Thus, a CreditRisk+ portfolio with a 99%-VaR of 5% and an average PD of 1% corresponds to an implied Vasicek asset correlation of

$$\rho \approx 11.1\%.$$

It is worth noting that the implied asset correlation obtained from the VaR calibration, is numerically close to the value obtained in the variance-matching example above. This similarity is accidental and should not be viewed as a general property of the two calibration approaches. Variance matching aligns the dispersion of the systematic factor, whereas VaR matching aligns a specific point of the loss distribution. In general, the two methods produce different values of ρ, particularly when the portfolio PD, confidence level, or dispersion of the CreditRisk+ factor are changed.

Effect of a non-unit LGD The derivation above assumed

$$\mathrm{LGD} = 1$$

to simplify the notation. For a constant LGD different from unity, the portfolio loss is

$$L(z) = \mathrm{LGD} \cdot p(z),$$

and the Vasicek 99% loss quantile becomes

$$q_V(0.99; \rho) = \mathrm{LGD} \cdot \Phi\left(\frac{\Phi^{-1}(p) + \sqrt{\rho}\, \Phi^{-1}(0.99)}{\sqrt{1-\rho}} \right).$$

The calibration condition is therefore

$$q_V(0.99; \rho) = q_{CR+}(0.99),$$

or equivalently

$$\Phi\left(\frac{\Phi^{-1}(p) + \sqrt{\rho}\, \Phi^{-1}(0.99)}{\sqrt{1-\rho}} \right) = \frac{q_{CR+}(0.99)}{\mathrm{LGD}}.$$

Hence, the calibration proceeds exactly as before after replacing the CreditRisk+ quantile by the normalized loss level

$$\frac{q_{CR+}(0.99)}{\mathrm{LGD}}.$$

In terms of the notation introduced above, the parameter

$$c = \Phi^{-1}(q_{CR+}(0.99))$$

is replaced by

$$c = \Phi^{-1}\left(\frac{q_{CR+}(0.99)}{\mathrm{LGD}} \right),$$

and the quadratic equation becomes

$$(b^2 + c^2)x^2 + 2ab\, x + (a^2 - c^2) = 0.$$

Thus, a constant LGD merely rescales the loss distribution and does not alter the structure of the calibration problem. The implied asset correlation is obtained in exactly the same way as in the unit-LGD case.

4.3 Matching Multiple Loss Quantiles

The single-quantile calibration described above provides an exact correspondence between the Vasicek and CreditRisk+ loss distributions at one selected confidence level. In practice, however, reliance on a single point in the tail may produce a calibration that fits well at one confidence level while exhibiting material deviations at other relevant loss levels. As throughout this section, the unconditional probability of default p is assumed to be known and identical in both frameworks. The CreditRisk+ model is treated as the reference model, and the objective is to determine the Vasicek asset correlation ρ that provides the closest overall match to the CreditRisk+ tail distribution. Let

$$\mathcal{A} = \{0.95,\, 0.99,\, 0.999\}$$

denote the set of confidence levels used for calibration. The asymptotic Vasicek quantile is

$$q_V(\alpha;\rho) = \Phi\left(\frac{\Phi^{-1}(p) + \sqrt{\rho}\,\Phi^{-1}(\alpha)}{\sqrt{1-\rho}}\right),$$

while the corresponding CreditRisk+ quantile is

$$q_{CR+}(\alpha) = p\, q_\Lambda(\alpha),$$

where q_Λ denotes the quantile function of the gamma-distributed systematic risk factor Λ.

Why exact matching is impossible. The Vasicek model contains a single free parameter governing tail dependence, namely the asset correlation ρ. Consequently, only one independent feature of the CreditRisk+ tail distribution can be matched exactly. Exact matching of several confidence levels would require

$$q_V(\alpha_i;\rho) = q_{CR+}(\alpha_i), \qquad i = 1, \ldots, m,$$

for $m > 1$. In general, no value of ρ will satisfy all equations simultaneously. The reason is that the two models generate their loss distributions through fundamentally different mechanisms. In the Vasicek framework, the systematic factor enters through the nonlinear probit transformation

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right),$$

whereas CreditRisk+ maps the systematic factor linearly through the stochastic intensity Λ. As a result, the shapes of the two quantile functions generally differ, making exact agreement across a range of confidence levels impossible.

Least-squares calibration. To obtain the closest overall fit, the Vasicek asset correlation is chosen to minimize the sum of squared deviations from the CreditRisk+ quantiles. Specifically,

$$\hat{\rho} = \arg\min_{\rho} \sum_{\alpha \in \mathcal{A}} w_\alpha \left[q_V(\alpha;\rho) - q_{CR+}(\alpha)\right]^2,$$

where $w_\alpha \geq 0$ are user-specified weights. Unless otherwise stated, equal weights are used,

$$w_{0.95} = w_{0.99} = w_{0.999} = 1.$$

This gives equal importance to the 95%, 99%, and 99.9% quantiles. Alternative weighting schemes may be employed if greater emphasis is desired on the extreme tail. For example, assigning a larger weight to the 99.9% quantile prioritizes agreement at confidence levels commonly used in economic-capital calculations. The resulting estimate $\hat{\rho}$ should therefore be interpreted as the Vasicek correlation that provides the best overall representation of the CreditRisk+ tail region rather than an exact point-by-point equivalence. Residual differences remain unavoidable and reflect the structural distinction between the nonlinear probit mapping in the Vasicek model and the linear factor representation used in CreditRisk+.

The three calibration approaches may therefore be viewed as increasing levels of approximation. Variance matching seeks agreement in the second moment of the systematic risk factor, single-quantile matching seeks exact agreement at one selected confidence level, while the multi-quantile approach seeks the Vasicek asset correlation that provides the best overall fit to several CreditRisk+ tail quantiles simultaneously. In an empirical analysis, the latter approach is adopted using the 95%, 99%, and 99.9% loss quantiles, as it provides a more balanced representation of the tail behaviour than calibration to a single point of the distribution.

5 Conclusion

The Vasicek model and CreditRisk+ provide alternative representations of systematic credit risk in large portfolios. Although their mathematical formulations differ, both models generate default dependence through exposure to a common systematic risk factor and produce tractable asymptotic loss distributions. Throughout the paper, the unconditional probability of default is assumed to be identical in both frameworks, while CreditRisk+ is treated as the reference model. The objective is therefore to determine the Vasicek asset correlation that reproduces selected characteristics of the CreditRisk+ loss distribution. Several calibration approaches have been considered. Matching the relative contributions of systematic and idiosyncratic risk leads to the approximate relationship

$$\rho \approx \frac{CV^2}{1 + CV^2},$$

which provides a simple translation between the Vasicek asset correlation and the coefficient of variation of the CreditRisk+ systematic risk factor. A more direct correspondence can be obtained by matching a specific loss quantile, yielding an implied asset correlation consistent with a chosen CreditRisk+ Value-at-Risk level. Because the two models employ fundamentally different mappings from the systematic factor to portfolio losses, exact agreement across the entire loss distribution is generally impossible. For this reason, a multi-quantile calibration based on the 95%, 99%, and 99.9% quantiles provides a natural compromise and allows the Vasicek model to reproduce the overall shape of the CreditRisk+ tail distribution as closely as possible. The results show that, despite their different theoretical foundations, the two frameworks can be calibrated to produce very similar loss distributions in large and well-diversified portfolios, particularly in the range of tail outcomes relevant for risk measurement, economic capital, and stress testing.